

EE2027 Cheat Sheet

Brians Tjipto

AC Circuit Elements (Phasors):

- **Capacitor:** $i(t) = C \frac{dv}{dt} \rightarrow Z_C = \frac{1}{j\omega C} = \frac{-j}{\omega C}$
- **Inductor:** $v(t) = L \frac{di}{dt} \rightarrow Z_L = j\omega L$
- **Superposition:** Kill Voltage Source $\rightarrow$ Short Circuit. Kill Current Source $\rightarrow$ Open Circuit.

Amplifier Analysis Steps

1. DC Analysis (Biasing):

- Open all capacitors ($C \rightarrow \infty$ impedance).
- Remove AC sources (Short AC Voltage, Open AC Current).
- Find I_C, V_{CE} (BJT) or I_D, V_{GS} (MOS).
- Calculate small signal params (g_m, r_π, r_o).
- **Check Mode:** Active (BJT) or Saturation (MOS).

2. AC Analysis (Small Signal):

- Short all DC Voltage sources ($V_{DD} \rightarrow GND$).
- Open all DC Current sources.
- Short all coupling/bypass capacitors.
- Replace Transistor with Small Signal Model (Hybrid- π or T-model).

Resistance Lookup Table

BJT Reflection Rules:

Looking Into	Resistance (R_{eq})
Base (Emitter Gnd)	r_π
Base (with R_E)	$r_\pi + (\beta + 1)R_E \approx r_\pi(1 + g_m R_E)$
Emitter (Base Gnd)	$r_e \approx 1/g_m$
Emitter (with R_S)	$\frac{r_\pi + R_S}{\beta + 1} \approx \frac{1}{g_m} + \frac{R_S}{\beta}$
Collector (Early Eff)	$r_o[1 + g_m R'_E]$ (Boosted r_o)

MOSFET Reflection Rules:

Looking Into	Resistance (R_{eq})
Gate	∞
Source (Gate Gnd)	$1/(g_m + g_{mb}) \approx 1/g_m$
Source (with R_G)	$\frac{1}{g_m}$ (Independent of R_G since $I_G = 0$)
Drain (Source Gnd)	r_o
Drain (with R_S)	$r_o[1 + (g_m + g_{mb})R_S]$

Mesh, Thevenin & Norton Methods

A. Mesh Analysis (Basic Steps)

- Assign mesh currents $I_1, I_2, \dots$ (clockwise by default).
- Apply KVL around each mesh: sum of voltage drops = sum of rises.
- Shared resistor between meshes: voltage drop is $R(I_1 - I_2)$.
- Solve simultaneous equations for all mesh currents.
- **Power check:** Power supplied = Power absorbed.

B. Thevenin Equivalent

Goal: Replace a circuit at terminals A–B with a single V_{th} source in series with R_{th} .

- **1. Find V_{th} (Open-Circuit Voltage):**
 - Remove the load.
 - Find the voltage across terminals A–B. This is $V_{OC} = V_{th}$.
- **2. Find R_{th} (Equivalent Resistance):**
 - Turn off all independent sources:
 - * Voltage source $\rightarrow$ short circuit.
 - * Current source $\rightarrow$ open circuit.
 - Look into A–B and compute the equivalent resistance.
- **(Alternative Method)** Use short-circuit current:

$$R_{th} = \frac{V_{OC}}{I_{SC}}$$

C. Norton Equivalent

Goal: Replace a circuit at terminals A–B with a current source I_N in parallel with R_N .

- **1. Find I_N (Short-Circuit Current):**
 - Short terminals A–B.
 - Compute the short-circuit current I_{SC} . This is I_N .
- **2. Find R_N :**
 - Same as Thevenin: turn off all independent sources.
 - Look into A–B for equivalent resistance.
 - Note: $R_N = R_{th}$ always.

- **3. Conversion:**

$$V_{th} = I_N R_N, \quad I_N = \frac{V_{th}}{R_{th}}$$

I. Basic Concepts & Power

Fundamental Constants:

- $V_T = kT/q \approx 25.85mV$ (use 25mV if specified)
- $k = 1.381 \times 10^{-23} J/K$, $q = 1.602 \times 10^{-19} C$
- $\epsilon_0 = 8.854 \times 10^{-14} F/cm$, $\epsilon_{Si} = 11.7\epsilon_0$, $\epsilon_{ox} = 3.9\epsilon_0$

Signals:

- **Average Value:** $V_{avg} = \frac{1}{T} \int_0^T v(t) dt$
- **RMS Value:** $V_{rms} = \sqrt{\frac{1}{T} \int_0^T v^2(t) dt}$ ($V_m/\sqrt{2}$ for sine)

Power Analysis:

- **Decibels (dB):**
 - Power: $10 \log_{10}(P_{out}/P_{in})$
 - Voltage: $20 \log_{10}(V_{out}/V_{in})$
- **dBm:** $P(dBm) = 10 \log_{10}(P_{out}/1mW)$
- **Complex Power:** $S = P + jQ = \mathbf{V}_{rms} \mathbf{I}_{rms}^*$

- **Real (Avg) Power:** $P = Re[VI^*] = V_{rms} I_{rms} \cos(\theta_v - \theta_i)$

- **Apparent Power:** $|S| = |V||I| = V_{rms} I_{rms}$

- **Power Factor:** $PF = \cos(\theta_i - \theta_v)$ (Leading if $\theta_i > \theta_v$)

- **Max Power Transfer:** Occurs when $R_{load} = R_{source}$

$$P_{max} = \frac{V_{th}^2}{4R_{th}}$$

II. Diode Circuits

Equation: $I_D = I_S(e^{\frac{V_D}{nV_T}} - 1)$
 $V_D = nV_T \ln\left(\frac{I_D}{I_S} + 1\right) \approx nV_T \ln\left(\frac{I_D}{I_S}\right)$

$$I_S = \frac{I_D}{e^{\frac{V_D}{nV_T}} - 1} \approx I_D e^{-\frac{V_D}{nV_T}}$$

Small-Signal Res: $r_d = \frac{nV_T}{I_D}$

DC Bias (KVL): $R = \frac{V_{DD} - V_D}{I_D}$

AC Analysis: Short DC sources ($V_{DD} \rightarrow 0$). $i_d = \frac{v_{ac}}{R + r_d}$

Breakdown: Diode clamps at $-V_Z$.

$$P_{npn} = V_{CE} I_C$$

2. PNP Transistor (Active Mode)

Current flows INTO Emitter. Emitter is higher than Base.

- **Basic KVL Loop:**

$$V_B = V_E - V_{EB}$$

- **With Emitter Resistor (R_E to V_{CC}):**

$$V_B = (V_{CC} - I_E R_E) - V_{EB}$$

3. Determining the “Diode Drop” (V_{BE} or V_{EB})

- **Standard Assumption:**

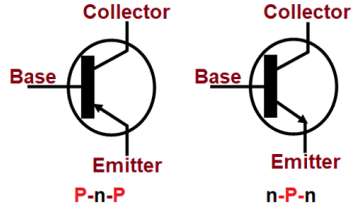
$$V_{BE} \approx 0.7 V \quad (\text{or } 0.6 V \text{ to } 0.7 V)$$

- **Exact Calculation (Inverted Shockley):**

$$V_{BE} = V_T \ln\left(\frac{I_C}{I_S}\right)$$

where $V_T \approx 0.025 V$ at $300K$.

III. BJT (NPN & PNP)



NPN Transistor (Active Mode)

Current flows OUT of Emitter. Base is higher than Emitter.

• Basic KVL Loop:

$$V_B = V_E + V_{BE}$$

• With Emitter Resistor (R_E to Ground):

$$V_B = (I_E R_E) + V_{BE}$$

• Substituted with I_C (useful for analysis):

$$V_B = \left[I_C \left(\frac{\beta + 1}{\beta} \right) R_E \right] + V_{BE}$$

• Current Source Biasing:

$$V_B = I_{R_B} \times R_B = (I_A - I_B)R_B, \quad I_O = I_C = \frac{\beta}{\beta + 1} I_E, \quad I_E = V/R_E$$

Large Signal (Active):

- $i_C = I_S e^{v_{BE}/V_T} (1 + v_{CE}/V_A)$ (NPN)
- $i_C = I_S e^{v_{EB}/V_T} (1 + v_{EC}/V_A)$ (PNP)
- $i_B = i_C/\beta$, $\beta = i_C/i_B$, $i_E = i_C + i_B = i_C/\alpha$
- $\alpha = \beta/(\beta + 1)$, $\beta = \alpha/(1 - \alpha)$
- $V_E = I_E R_E = \left(\frac{\beta + 1}{\beta} \right) I_C R_E \approx I_C R_E$ (since $\frac{\beta + 1}{\beta} \approx 1$)

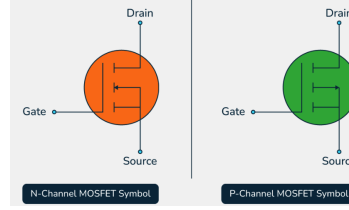
Approximation Notes (I_C Calculation):

- **Why remove the “-1”?** The full term is $(e^{V_{BE}/V_T} - 1)$. Since $V_{BE} \gg V_T$ (e.g., $0.64V \gg 0.025V$), the exponential term is massive ($\approx 10^{11}$). Subtracting 1 is negligible.
- **Why remove Early Effect?** If V_A is not provided, assume $V_A = \infty$. The term $(1 + V_{CE}/V_A)$ becomes 1.
- **Result:** Use the simplified form $I_C \approx I_S e^{V_{BE}/V_T}$.

Small-Signal Parameters:

- $g_m (mA/V) = I_C/V_T$
 g_m : transconductance; small-signal gain = $\Delta I_C/\Delta V_{BE}$, showing how V_{BE} controls I_C .
- $r_\pi = v_{be}/i_b = \beta/g_m = \beta V_T/I_C$
- $r_o = v_{ce}/i_c = V_A/I_C$

IV. MOSFET (NMOS & PMOS)



Physical Parameters:

- $C_{ox} = \epsilon_{ox}/t_{ox}$ (Gate capacitance/area)
- $K_n = \frac{1}{2} \mu_n C_{ox} (W/L)$, $K_p = \frac{1}{2} \mu_p C_{ox} (W/L)$

DC Operation (Large Signal):

- NMOS (Active/Sat when $V_{DS} \geq V_{GS} - V_{TH}$):

$$I_D = K_n (V_{GS} - V_{TH})^2 (1 + \lambda V_{DS})$$

- PMOS (Active/Sat when $V_{SD} \geq V_{SG} - |V_{TH}|$):

$$I_D = K_p (V_{SG} - |V_{TH}|)^2 (1 + \lambda V_{SD})$$

- **Channel Length Modulation (λ):** Used when I_D changes with V_{DS} .

$$\lambda = \frac{1}{V_A} \approx \frac{I_{D2} - I_{D1}}{I_{D1} V_{DS2} - I_{D2} V_{DS1}}$$

Voltage Relationships (KVL Loops)

1. NMOS Analysis (Current flows Drain $\rightarrow$ Source)

- **Drain Voltage (V_D):**
Usually connected to V_{DD} via R_D (or R_3).

$$V_D = V_{DD} - I_D R_D$$

- **Source Voltage (V_S):**
Usually connected to Ground via R_S (or R_4).

$$V_S = I_D R_S \quad (\text{If grounded directly, } V_S = 0)$$

• Drain-Source Voltage (V_{DS}):

$$V_{DS} = V_D - V_S = (V_{DD} - I_D R_D) - (I_D R_S)$$

• Feedback Case (Like Q8):

If Gate is connected to Drain via R_G , since $I_G = 0$:

$$V_G = V_D \implies V_{GS} = V_{DS}$$

2. PMOS Analysis (Current flows Source $\rightarrow$ Drain)

- **Source Voltage (V_S):**
Source is at the “top”. Connected to V_{DD} via R_S .

$$V_S = V_{DD} - I_D R_S \quad (\text{If no } R_S, V_S = V_{DD})$$

• Drain Voltage (V_D):

Drain is at the “bottom”. Connected to Ground via R_D .

$$V_D = I_D R_D$$

• Source-Drain Voltage (V_{SD}):

$$V_{SD} = V_S - V_D = (V_{DD} - I_D R_S) - (I_D R_D)$$

3. Gate Voltage (V_G) - Divider Rule

- If Gate is biased by two resistors R_1 (top) and R_2 (bottom):

$$V_G = V_{DD} \left(\frac{R_2}{R_1 + R_2} \right)$$

Note: If $I_G \neq 0$ is impossible in MOSFET DC, always assume $I_G = 0$.

Checking Region of Operation:

• 1. Channel Formed?

$$\text{NMOS: } V_{GS} > V_{TH} \quad \text{PMOS: } V_{SG} > |V_{TH}|$$

• 2. Saturation (Amplifier Mode)?

$$\text{NMOS: } V_{DS} \geq V_{GS} - V_{TH} \\ \text{PMOS: } V_{SD} \geq V_{SG} - |V_{TH}|$$

• 3. Triode (Linear)?

$$\text{NMOS: } V_{DS} < V_{GS} - V_{TH}$$

Small-Signal Parameters:

- $g_m = \frac{2I_D}{V_{GS} - V_{TH}} = \sqrt{4K_n I_D}$
(Transconductance: How V_{GS} controls current)
- $r_o = \frac{1}{\lambda I_D}$
(Output Resistance: Due to channel length mod.)
- **Body Effect:** If $V_{Source} \neq V_{Body}$, V_{TH} changes.
 $V_{TH_new} = V_{TH0} + \gamma(\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F})$

Equations (NMOS):

- **Linear:** $v_{DS} \leq v_{GS} - V_{TH}$

$$i_D = 2K_n [(v_{GS} - V_{TH})v_{DS} - \frac{1}{2}v_{DS}^2]$$

- **Linear Resistance (R_{DS}):** Small v_{DS} :

$$R_{DS} = \frac{1}{2K_n (v_{GS} - V_{TH})}$$

- **Saturation:** $v_{DS} \geq v_{DSsat} = v_{GS} - V_{TH}$

$$i_D = K_n (v_{GS} - V_{TH})^2 (1 + \lambda v_{DS})$$

PMOS: Use $|v_{GS}|, |V_{TH}|, |v_{DS}|$. Sat if $|v_{DS}| \geq |v_{GS}| - |V_{TH}|$.

Small-Signal Parameters:

- $g_m = 2K_n (V_{GS} - V_{TH}) = 2\sqrt{K_n I_D} = 2I_D/(V_{GS} - V_{TH})$
- $r_o = 1/(\lambda I_D) = V_A/I_D$
- $g_{mb} = \chi g_m$ (Body Effect)

V. Op-Amp & Filters

Basic Amps:

- Inverting: $-R_2/R_1$
- Non-Inv: $1 + R_2/R_1$
- Summing: $v_{out} = -(\frac{R_f}{R_1} v_1 + \frac{R_f}{R_2} v_2 + \dots)$
- Integrator: $-1/(sCR)$ — Differentiator: $-sCR$

Op-Amp Imperfections (Non-Ideal)

1. Gain-Bandwidth Product (GBW):

- Open Loop: $A(j\omega) \approx \frac{GBW}{j\omega}$
- Closed Loop BW ($f_{3dB,CL}$):

$$f_{3dB,CL} = GBW \times \beta = GBW \times \frac{R_1}{R_1 + R_2}$$

- Valid when $f \gg f_{3dB,OL}$.

2. Input Offset Voltage (V_{OS}): Models mismatch in differential pair. Modeled as a DC source at positive terminal.

$$V_{out,total} = \text{Ideal } V_{out} + \underbrace{\left(1 + \frac{R_2}{R_1}\right) V_{OS}}_{\text{DC Error Shift}}$$

3. Slew Rate (SR): Max rate of change of output voltage.

$$SR = \left. \frac{dV_{out}}{dt} \right|_{max} \implies \text{To avoid distortion:}$$

$$SR \geq 2\pi f_{max} V_{peak}$$

First Order Active Filters:

- **Low Pass:** $H(j\omega) = -\frac{R_2}{R_1} \frac{1}{1+j\omega R_2 C}$
 $\omega_c = \frac{1}{R_2 C}$, DC Gain = $-R_2/R_1$
- **High Pass:** $H(j\omega) = -\frac{R_2}{R_1} \frac{j\omega R_1 C}{1+j\omega R_1 C}$
 $\omega_c = \frac{1}{R_1 C}$, High Freq Gain = $-R_2/R_1$

Advanced Op-Amp Circuits:

- **Instrumentation Amp (2-OpAmp):**

$$v_{out} = -\frac{R_2}{R_1} (v_{id})$$

- **Log Amplifier:** $v_{out} = -V_T \ln\left(\frac{v_{in}}{R_1 I_S}\right)$
- **Exponential Amp:** $v_{out} = -R_1 I_S e^{v_{in}/V_T}$

Sallen-Key Filters (2nd Order):

- **SK Lowpass:**

$$\frac{v_{out}}{v_{in}} = \frac{H_o \omega_o^2}{s^2 + \frac{\omega_o}{Q} s + \omega_o^2}$$

Design: $\omega_o = \frac{1}{RC}$, $K = H_o$, $1/Q = 3 - K$.

- **SK Highpass:**

$$\frac{v_{out}}{v_{in}} = \frac{H_o s^2}{s^2 + \frac{\omega_o}{Q} s + \omega_o^2}$$

Design: $\omega_o = \frac{1}{RC}$, $K = H_o$, $1/Q = 3 - K$.

Application: BJT: high freq, high speed, high power circuits
MOSFET: Logic, memory, mixed signal circuits, low power, and VLSI Circuit

VI. Impedance Reflection (r_x)

BJT Equivalent Resistances:

- **Into Base (R_E at emitter):** $r_x = r_\pi + (\beta + 1)R_E \approx r_\pi(1 + g_m R_E)$
- **Into Emitter (R_S at base):** $r_x = \frac{R_S + r_\pi}{\beta + 1} \approx \frac{1}{g_m} + \frac{R_S}{\beta + 1}$
- **Into Collector (R_E at emitter):**
 $r_x \approx r_o[1 + g_m(R_E // (r_\pi + R_S)) \frac{r_\pi}{r_\pi + R_S}]$ Simplified ($R_S = 0$):
 $r_x \approx r_o(\beta + 1)$

MOSFET Equivalent Resistances:

- **Into Gate:** $r_x = \infty$
- **Into Source (R_G at gate):** $r_x = \frac{1}{g_m + g_{mb}}$ (if R_G ignored/small)
- **Into Drain (R_E at source):** $r_x = r_o[1 + (g_m + g_{mb})R_E]$

VII. Single Stage Configurations

BJT (Table 3):

Config	G_m	$A_v = v_{out}/v_{in}$
CE	g_m	$-g_m(R_{out} // R_L)$
CB	$-g_m$	$g_m(R_{out} // R_L)$
CC	N/A	$\frac{g_m R_L}{1 + g_m R_L}$
CE+Deg	$\frac{g_m}{1 + g_m R_E}$	$\frac{-g_m(R_{out} // R_L)}{1 + g_m R_E} \approx \frac{-(R_{out} // R_L)}{R_E}$

Degeneration Summary (R_E or R_S present):

- **Gain Reduction:** $A_v \approx \frac{-R_C}{R_E}$ (BJT) or $\frac{-R_D}{R_S}$ (MOS).
- **Linearity:** Increases linearity but sacrifices gain.
- **Bandwidth:** Increases bandwidth.
- **Input R (BJT):** Increases significantly ($R_{in} \approx \beta R_E$).

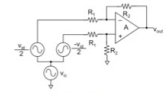
MOSFET (Table 4):

Config	G_m	$A_v = v_{out}/v_{in}$
CS	g_m	$-g_m(R_{out} // R_L)$
CG	$-(g_m + g_{mb})$	$(g_m + g_{mb})(R_{out} // R_L)$
CD	N/A	$\frac{g_m R_L}{1 + (g_m + g_{mb})R_L}$
CS+Deg	$\frac{g_m}{1 + (g_m + g_{mb})R_E}$	$\frac{-g_m(R_{out} // R_L)}{1 + (g_m + g_{mb})R_E}$

VIII. Op-Amp Specs & Noise

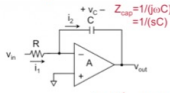
- **CMRR:** $20 \log(A_{OL}/A_{CM})$
- **PSRR:** $20 \log(A_{OL}/A_{SUP})$
- **Slew Rate:** $SR \geq 2\pi f_{max} V_{pk}$
- **GBW:** $f_{3dB,CL} \times A_{CL} \approx GBW$
- **Noise RMS:** $v_{noise,rms} = \sqrt{V_n^2 \times f_{3dB} \times \pi/2}$

7. Instrumentation.



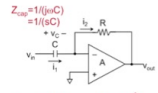
$V_{out} = \frac{R_2}{R_1} (V_{id})$
Rejects common mode.
Amplifies differential.
Good for blocking noise
Used in measuring instrument

8. Integrator



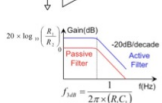
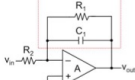
$V_{out}(t) = -\frac{1}{RC} \int_0^t V_{in}(t) dt$
 $H(j\omega) = -\frac{Z_{out}}{R}$
Gain $\downarrow$ as Freq $\uparrow$
in $\rightarrow$

9. Differentiator

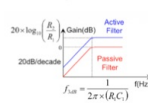
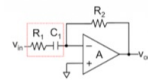


$V_{out}(t) = -RC \frac{d}{dt} V_{in}(t)$
 $Z_{out} = 1/(j\omega C)$
 $Z_{in} = 1/(j\omega C)$

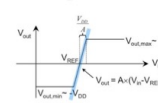
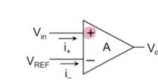
10. 1st Order Lowpass



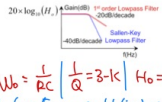
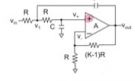
11. 1st Order High pass



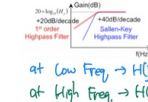
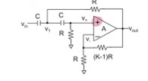
12. Comparator



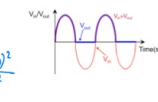
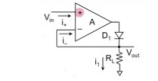
13. 2nd Order Sallen-Key Lowpass Filter



14. 2nd Order Sallen-Key High Pass Filter



15. Superdiode Rectifier

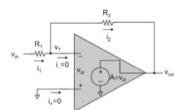


$\omega_o = \frac{1}{RC} \left[\frac{1}{Q} = 3 - K \right] H_o = K$
at Low Freq $\rightarrow H(j\omega) \approx H_o$
at High Freq $\rightarrow H(j\omega) \approx \frac{H_o \omega_o^2}{(j\omega)^2}$

at Low Freq $\rightarrow H(j\omega) \approx \frac{H_o \omega_o^2}{\omega^2}$
at High Freq $\rightarrow H(j\omega) \approx H_o$

Conf	r_x	Conf	r_x	Conf	r_x
	$r_x + (1 + \beta)R_E$ $\approx r_x(1 + g_m R_E)$ If $R_E = 0$ $r_x = r_\pi$		$r_x \left\{ 1 + g_m \left[(r_x + R_S) // R_E \right] \left(\frac{r_x}{r_x + R_S} \right) \right\}$ If $R_S = 0$ and $r_x \ll R_E$ $\Rightarrow r_{x,max} = r_x(\beta + 1)$ If $R_E = 0$, $r_x = r_o$		$\frac{1}{g_m}$
	$\frac{R_S + r_x}{1 + \beta}$ $\approx \frac{R_S}{1 + \beta} + \frac{1}{g_m}$ If $R_E = 0$ $r_x \approx \frac{1}{g_m}$		$\frac{1}{g_m} \times \frac{r_x + R_C}{r_x + R_C / \beta}$ If $R_C = 0$, $r_x \approx \frac{1}{g_m}$		

OP-AMPS



In circuit Analysis
 $R_{in} = \infty$
 $R_{out} = 0$
 $A = \infty$
 $i_- = 0$
 $i_+ = 0$
 $i_- = i_+$
 $V_- = V_+$ Virtual short

Open Loop (SSA)
 - Difficult to control gain
 - simple
 - ↓ Power consumption
 - Poor Linearity / ↑ Distortion

Feedback (Op-Amp) Amplifiers.
 - Accurate gain from Resistor
 - Complicated
 - ↑ Power consumption
 - High Linearity / ↓ Distortion

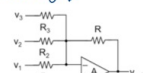
vs

1. Inverting



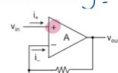
$$V_{out} = -\frac{R_2}{R_1} V_{in}$$

2. Summing



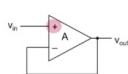
Superposition
 $V_{out} = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right)$

3. Non-inverting



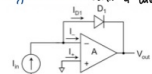
$$V_{out} = \left(1 + \frac{R_2}{R_1} \right) V_{in}$$

4. Source Follower



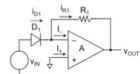
$$V_{out} = V_{in}$$

5. Logarithmic



$$V_{out} = -V_T \ln \left(\frac{I_{in}}{I_s} \right)$$

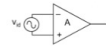
6. Exponential



$$V_{out} = -R_1 I_s e^{V_{in}/V_T}$$

Op Amp Parameters

Open Loop Voltage Gain (A_{OL})



How well the desired signal is amplified

Common Mode Voltage Gain (A_{CM})



How much noise is amplified Ideal Value is 0

Common Mode Rejection Ratio (CMRR)

$$CMRR [dB] = 20 \log \left(\frac{A_{OL}}{A_{CM}} \right)$$

How well noise is rejected Ideal is ∞

Input Offset Voltage (V_{OS})



DC output shift
 Use super-position to find a new V_{out}

Power Supply Rejection Ratio (PSRR)



How well supply line ripples are rejected
 Ideal is ∞

$$A_{sup} = \frac{V_{O,sup}}{V_{i,sup}}$$

$$PSRR [dB] = 20 \log \left(\frac{A_{OL}}{A_{sup}} \right)$$

$$GBW$$

$$f_{slew, cl} = 2\pi GBW \times \frac{R_1}{R_1 + R_2}$$

$$f_{slew, ac} = GBW \div A_{OL}$$

Slew Rate (SR)

$$SR = 2\pi f \cdot V_{out, pk} \cdot \sin(2\pi ft)$$

Maximum slope that can occur at output.

MOS	G_m	A_v
CS (A)	g_m	Derive Based on 2-port Network
CG (B)	$-(g_m - g_{mb})$ Drop g_{mb} if no body effect	Derive Based on 2-port Network
CD (C)	Too Complex To Be Useful	$\frac{g_m R_L}{1 + (g_m - g_{mb}) R_L} \approx \frac{g_m}{g_m - g_{mb}}$ Drop g_{mb} if no body effect
CS with R_E (D)	$\frac{g_m}{1 + (g_m - g_{mb}) R_E}$ Drop g_{mb} if no body effect	Derive Based on 2-port Network

 Input - Base Output - Collector Neither - Emitter ⇒ CE	 Input - Emitter Output - Collector Neither - Base ⇒ CB	 Input - Base Output - Emitter Neither - Collector ⇒ CC	 Input - Base Output - Collector Neither - Emitter (with resistor) ⇒ CE with degeneration
 Input - Gate Output - Drain Neither - Source ⇒ CS	 Input - Source Output - Drain Neither - Gate ⇒ CG	 Input - Gate Output - Source Neither - Drain ⇒ CD	 Input - Gate Output - Drain Neither - Source (with resistor) ⇒ CS with degeneration

Found: <https://github.com/brianstm/NUS.git>

Conf	r_x	Conf	r_x	Conf	r_x
(A)		(B)	$r_x [1 + (g_m - g_{mb}) R_E]$ If $R_E = 0, r_x = r_o$	(E)	$\frac{1}{g_m}$
(C)		(D)	$\frac{1}{g_m - g_{mb}} \times \frac{r_o + R_C}{r_o}$ If $R_C = 0, r_x \approx \frac{1}{g_m - g_{mb}}$		

BJT	G_m	A_v
CE (A)	g_m	Derive Based on 2-port Network
CB (B)	$-g_m$	Derive Based on 2-port Network
CC (C)	Too Complex To Be Useful	$\frac{g_m R_L}{1 + g_m R_L}$
CE with Emitter Degeneration (D)	$\frac{g_m}{1 + g_m R_E}$	Derive Based on 2-port Network